Circuit model and solder practice

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Test a circuit representation before making permanent connections.

SUCCESS CRITERIA - I can...

1. Build a complete battery-switch-motor circuit model.
2. Explain an open versus closed path.
3. Document a supervised solder-practice check.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “open circuit” and “short circuit” to explain your model or result. Sentence starters are optional.

open circuit: a break that prevents current through that path

short circuit: an unintended low-resistance path bypassing a component

insulation: material that limits unwanted electric conduction

01 Warm-up: bring back an earlier idea

A car prototype is ready for a planned 5 m powered run. **Name** a safety check before power and a measurement needed to evaluate the run.

Word help: evaluate = check a claim against evidence and conditions

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $I = \Delta q/\Delta t$; $I = 0$ in an ideal open circuit.
Use the assigned simulation and teacher solder procedure. Disconnect power; hot irons/joints stay in the designated work area.

Worked reasoning: In a simple car circuit, battery positive connects through the switch and motor back to battery negative. A wire directly across the battery bypasses the motor and can create a hazardous short circuit.

Your turn: Draw the intended closed loop and circle a place where an open switch safely breaks it.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw your simulated closed circuit and an open-switch version.
Predict which runs the motor and why.

Sentence starter: I predict … . My model supports this because … .



Circuit symbols: long/short lines = battery; rectangles = resistors; M = motor.

Marzano check - circle: 1 2 3

04 Apply the physics

In a circuit simulation, a 3 V supply delivers 0.20 A to a motor.
Label voltage and current and calculate input power. Make one approved change in the simulation and record what you actually observe.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Make a two-column record: practice-joint observation and next improvement. Include teacher feedback on the joint and tool handling.

Sentence starter: I notice … . This supports … because … .

Observed joint / handling	Teacher feedback	Next time
___	___	___
___	___	___

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Parallel branches draw 0.4 A and 0.6 A.
Find the current supplied to their junction.
Draw the stated circuit and label quantities relevant to the question.

Word help: current = rate of charge flow

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

Marzano check - circle: 1 2 3

07 Exit check

Before soldering a circuit connection, should power be connected or disconnected?
State one reason.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw two circuit faults on the blank models: (A) one disconnected lead; (B) a wire that bypasses the motor and connects across the battery.
Explain the expected behavior and identify a teacher-approved first check with power disconnected. These are drawings, not faults to build.

Sentence starter: I predict … . My model supports this because … .

A: open lead

B: bypass

Draw each circuit fault; label the motor and battery.

Marzano check - circle: 1 2 3